A Twin-Prime Propagation Conjecture

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Abstract

I introduce a natural generative process on consecutive lower twin primes. Let p_n denote the nth lower twin prime. Define

$$C_n = p_n + p_{n+1} + 1, \qquad D_n = p_n + p_{n+1} + 3.$$

I say that the pair (p_n, p_{n+1}) *propagates* if at least one of C_n or D_n is itself a lower twin prime. Extensive computation up to 10^{13} shows that the success rate of this process declines slowly and regularly. I conjecture that infinitely many pairs propagate. A conditional argument under the Hardy–Littlewood prime tuples conjecture is outlined.

1 Introduction

Let $p_1 < p_2 < p_3 < \dots$ be the sequence of lower members of twin primes (i.e., both p_n and $p_n + 2$ are prime). For each $n \geq 1$ I form the two linear expressions

$$C_n = p_n + p_{n+1} + 1, \qquad D_n = p_n + p_{n+1} + 3. \tag{1}$$

I say that the ordered pair (p_n, p_{n+1}) **propagates** if at least one of C_n or D_n is again the lower member of a twin prime.

Conjecture 1 (Twin-Prime Propagation Conjecture)**.** *There are infinitely many indices n for which (p_n, p_{n+1}) propagates.*

2 Computational Evidence

The conjecture has been tested for all lower twin primes $p_n \leq 10^{13}$. In this range the empirical success rate declines smoothly. In the regime $p_n \geq 2^{20}$ the rate is very well approximated by the model

$$\approx \frac{6.175}{(\ln p_n)^{2.657}\,(\ln \ln p_n)^{-2.258}}, \tag{2}$$

which achieves a coefficient of determination $R^2 = 0.99858$.

The absolute number of successful propagations continues to increase throughout every complete dyadic interval examined, consistent with the conjectured infinitude.

3 Conditional Argument

Assume a sufficiently strong form of the Hardy–Littlewood prime tuples conjecture. Then the probability that an integer of size x is a lower twin prime is asymptotically

$$\sim 2C_2 \frac{1}{(\log x)^2},$$

where C_2 is the twin-prime constant. For $p_n \approx x$ both C_n and D_n are of size $\sim 2x$. Consequently the probability of a successful propagation at height x is

$$\asymp \frac{1}{(\log x)^2}.$$

Summing over the lower twins themselves yields an expected number of successes up to X of order

$$\int_2^X \frac{dt}{(\log t)^3},$$

which diverges as $X \to \infty$. Hence, under the Hardy–Littlewood conjecture, there are infinitely many successful propagations.

4 Remarks

The same heuristic applies, with only minor changes, if one weakens the condition on C_n or D_n from "lower twin prime" to "prime". The computational data are fully consistent with the asymptotic density predicted by the Hardy–Littlewood heuristics.

References

[1] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio Numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.